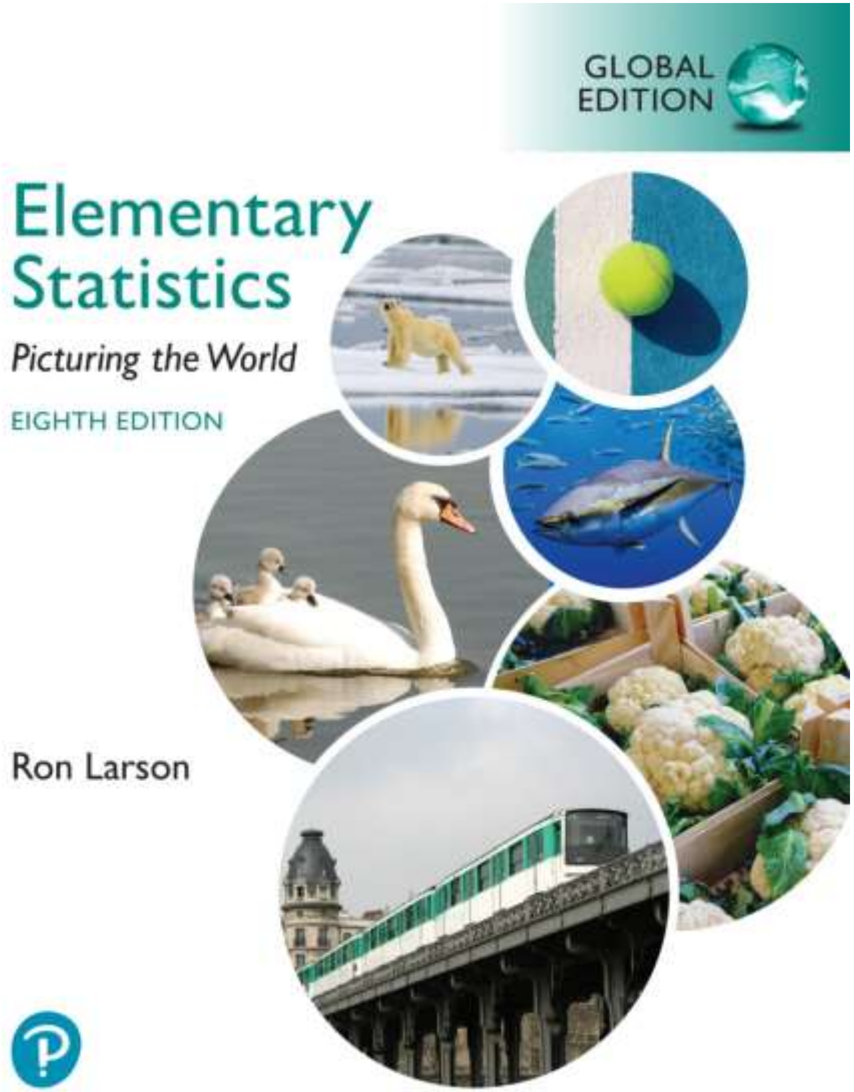
Chapter 2

Descriptive Statistics



Chapter Outline

- 2.1 Frequency Distributions and Their Graphs
- 2.2 More Graphs and Displays
- 2.3 Measures of Central Tendency
- 2.4 Measures of Variation
- 2.5 Measures of Position

Section 2.2

More Graphs and Displays

Section 2.2 Objectives

- How to graph and interpret quantitative data using stem-and-leaf plots and dot plots
- How to graph and interpret qualitative data using pie charts and Pareto charts
- How to graph and interpret paired data sets using scatter plots and time series charts

Graphing Quantitative Data Sets

Stem-and-leaf plot

- Each number is separated into a **stem** and a **leaf**.
- Similar to a histogram.
- Still contains original data values.
- Provides an easy way to sort data.

26

2	1 5 5 6 7 8
3	0 6 6
4	5

Data: 21, 25, 25, **26**, 27, 28,
30, 36, 36, 45

Example: Constructing a Stem-and-Leaf Plot

The data set lists the numbers of text messages sent in one day by 50 cell phone users. Display the data in a stem-and-leaf plot. Describe any patterns.

Number of Text Messages Sent				
75	49	104	59	88
123	75	109	68	81
66	80	78	69	55
76	114	98	73	18
42	84	46	52	25
25	26	33	25	20
32	24	43	17	49
27	32	29	29	40
23	33	30	41	35
38	36	54	30	148

Solution: Constructing a Stem-and-Leaf Plot

- The data entries go from a low of 17 to a high of 148.
- Use the rightmost digit as the leaf.
  - For instance,
    76 = 7 | 6 and
    104 = 10 | 4
- List the stems, 7 to 14, to the left of a vertical line.
- For each data entry, list a leaf to the right of its stem.

Number of Text Messages Sent				
75	49	104	59	88
123	75	109	68	81
66	80	78	69	55
76	114	98	73	18
42	84	46	52	25
25	26	33	25	20
32	24	43	17	49
27	32	29	29	40
23	33	30	41	35
38	36	54	30	148

Solution: Constructing a Stem-and-Leaf Plot

Number of Text Messages Sent

Key: 10|4 = 104

1	7 8
2	0 3 4 5 5 5 6 7 9 9
3	0 0 2 2 3 3 5 6 8
4	0 1 2 3 6 9 9
5	2 4 5 9
6	6 8 9
7	3 5 5 6 8
8	0 1 4 8
9	8
10	4 9
11	4
12	3
13	
14	8

From the display, you can see that more than 50% of the cell phone users sent between 20 and 50 text messages.

Example: Constructing Variations of Stem-and-Leaf Plots

Organize the data set in previous example using a stem-and-leaf plot that has two rows for each stem. Describe any patterns.

Number of Text Messages Sent				
75	49	104	59	88
123	75	109	68	81
66	80	78	69	55
76	114	98	73	18
42	84	46	52	25
25	26	33	25	20
32	24	43	17	49
27	32	29	29	40
23	33	30	41	35
38	36	54	30	148

Solution: Constructing Variations of Stem-and-Leaf Plots

Number of Text Messages Sent

Key: 10|4 = 104

1	
1	7 8
2	0 3 4
2	5 5 5 6 7 9 9
3	0 0 2 2 3 3
3	5 6 8
4	0 1 2 3
4	6 9 9
5	2 4
5	5 9
6	
6	6 8 9
7	3
7	5 5 6 8
8	0 1 4
8	8
9	
9	8
10	4
10	9
11	4
11	
12	3
12	
13	
13	
14	
14	8

- List each stem twice.
- Use the leaves 0, 1, 2, 3, and 4 in the first stem row and the leaves 5, 6, 7, 8, and 9 in the second stem row.
- Notice that by using two rows per stem, you obtain a more detailed picture of the data.

Solution: Constructing Variations of Stem-and-Leaf Plots

Number of Text Messages Sent

Key: 10|4 = 104

1	
1	7 8
2	0 3 4
2	5 5 5 6 7 9 9
3	0 0 2 2 3 3
3	5 6 8
4	0 1 2 3
4	6 9 9
5	2 4
5	5 9
6	
6	6 8 9
7	3
7	5 5 6 8
8	0 1 4
8	8
9	
9	8
10	4
10	9
11	4
11	
12	3
12	
13	
13	
14	
14	8

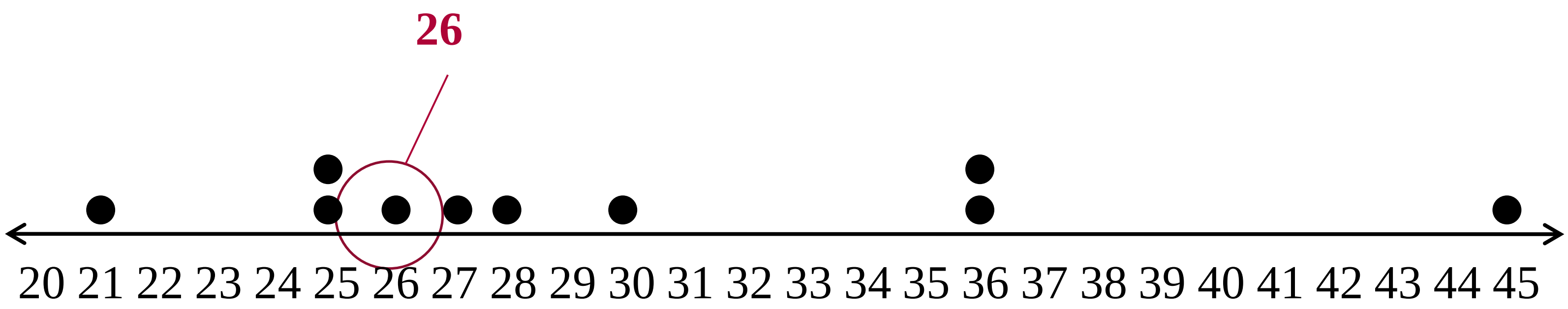
From the display, you can see that most of the cell phone users sent between 20 and 80 text messages.

Graphing Quantitative Data Sets

Dot plot

- Each data entry is plotted, using a point, above a horizontal axis

Data: 21, 25, 25, **26**, 27, 28, 30, 36, 36, 45



Example: Constructing a Dot Plot

Use a dot plot to organize the data set in Example 1. Describe any patterns.

Number of Text Messages Sent									
75	49	104	59	88	123	75	109	68	81
66	80	78	69	55	76	114	98	73	18
42	84	46	52	25	25	26	33	25	20
32	24	43	17	49	27	32	29	29	40
23	33	30	41	35	38	36	54	30	148

Solution: Constructing a Dot Plot

Number of Text Messages Sent									
75	49	104	59	88	123	75	109	68	81
66	80	78	69	55	76	114	98	73	18
42	84	46	52	25	25	26	33	25	20
32	24	43	17	49	27	32	29	29	40
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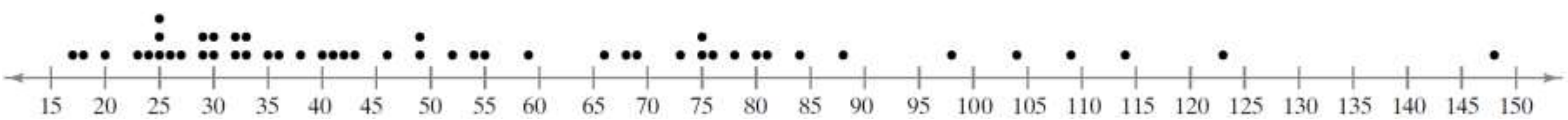
Number of Text Messages Sent

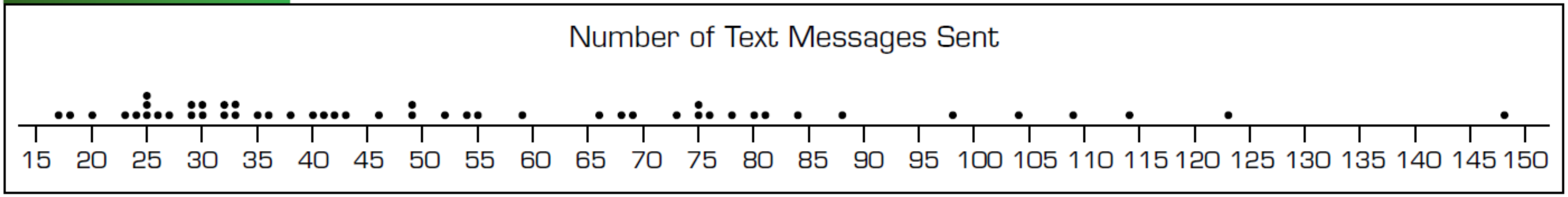


From the dot plot, you can see that most entries occur between 20 and 80 and only 4 people sent more than 100 text messages. You can also see that 148 is an unusual data entry.

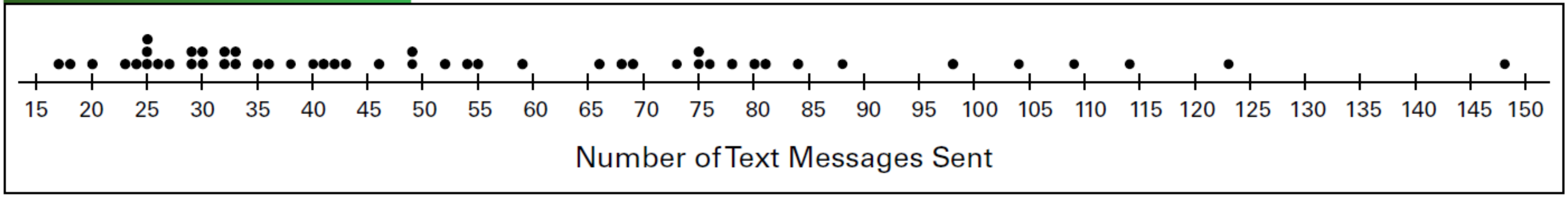
Solution: Constructing a Dot Plot

Technology can be used to construct dot plots. For instance, Minitab and StatCrunch dot plots for the text messaging data are shown below.

MINITAB



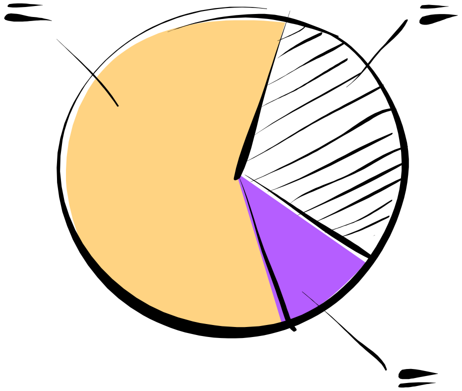
STATCRUNCH



Graphing Qualitative Data Sets

Pie Chart

- Pie charts provide a convenient way to present qualitative data graphically as percents of a whole.
- A circle is divided into sectors that represent categories.
- The area of each sector is proportional to the frequency of each category.



Example: Constructing a Pie Chart

The numbers of earned degrees conferred (in thousands) in 2019 are shown in the table. Use a pie chart to organize the data. ***(Source: U.S. National Center for Educational Statistics)***

Earned Degrees Conferred in 2019

Type of degree	Number (in thousands)
Associate's	1037
Bachelor's	2013
Master's	834
Doctoral	188

Solution: Constructing a Pie Chart

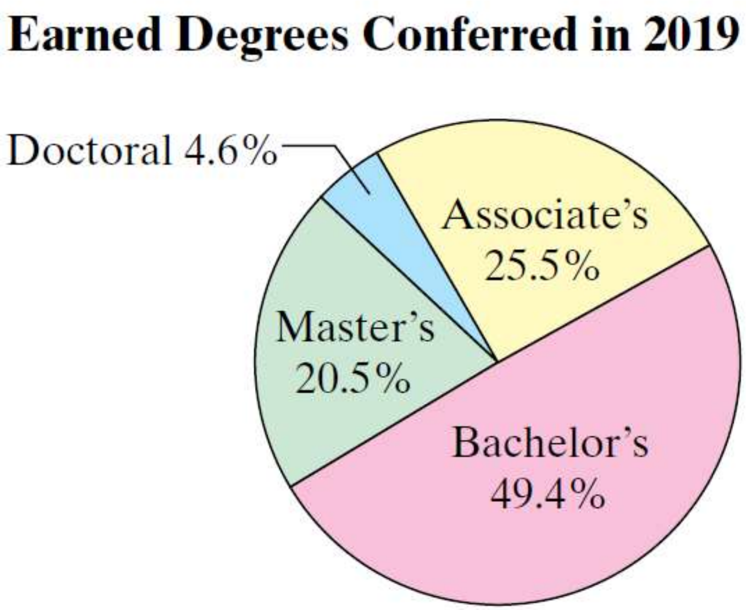
- Construct the pie chart using the central angle that corresponds to each category.
  - To find the central angle, multiply 360º by the category's relative frequency.
  - For example, the central angle for associate's degree is

$$360^{\circ}\,(0.255) \approx 91.8^{\circ}$$

Solution: Constructing a Pie Chart

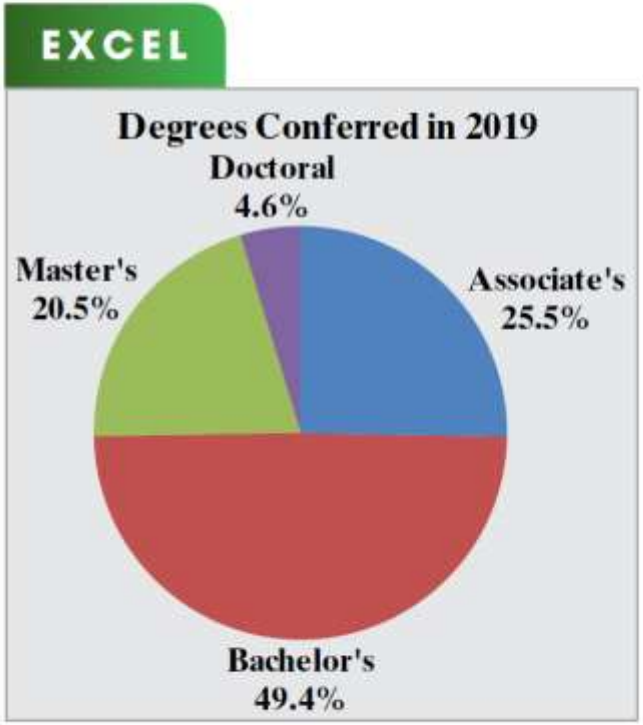
- Find the relative frequency (percent) of each category.

Type of degree	f	Relative frequency	Angle
Associate's	1037	0.255	91.8°
Bachelor's	2013	0.494	177.8°
Master's	834	0.205	73.8°
Doctoral	188	0.046	16.6°



Solution: Constructing a Pie Chart

Check using technology

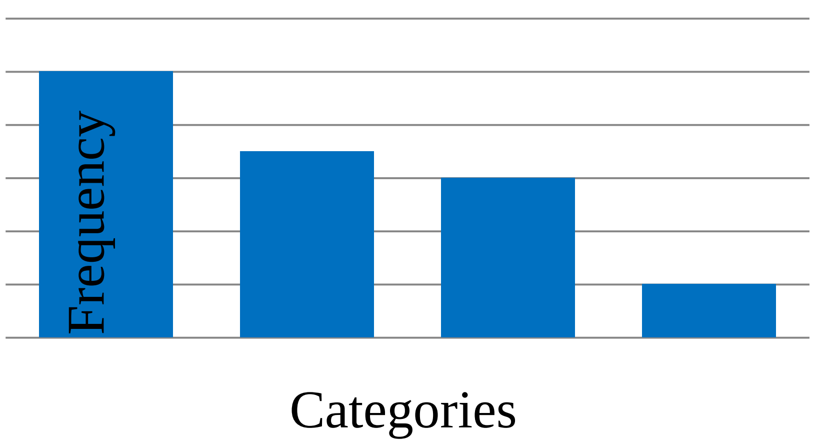


From the pie chart, you can see that almost one-half of the degrees conferred in 2019 were bachelor's degrees.

Graphing Qualitative Data Sets

Pareto Chart

- A vertical bar graph in which the height of each bar represents frequency or relative frequency.
- The bars are positioned in order of decreasing height, with the tallest bar positioned at the left.



Example: Constructing a Pareto Chart

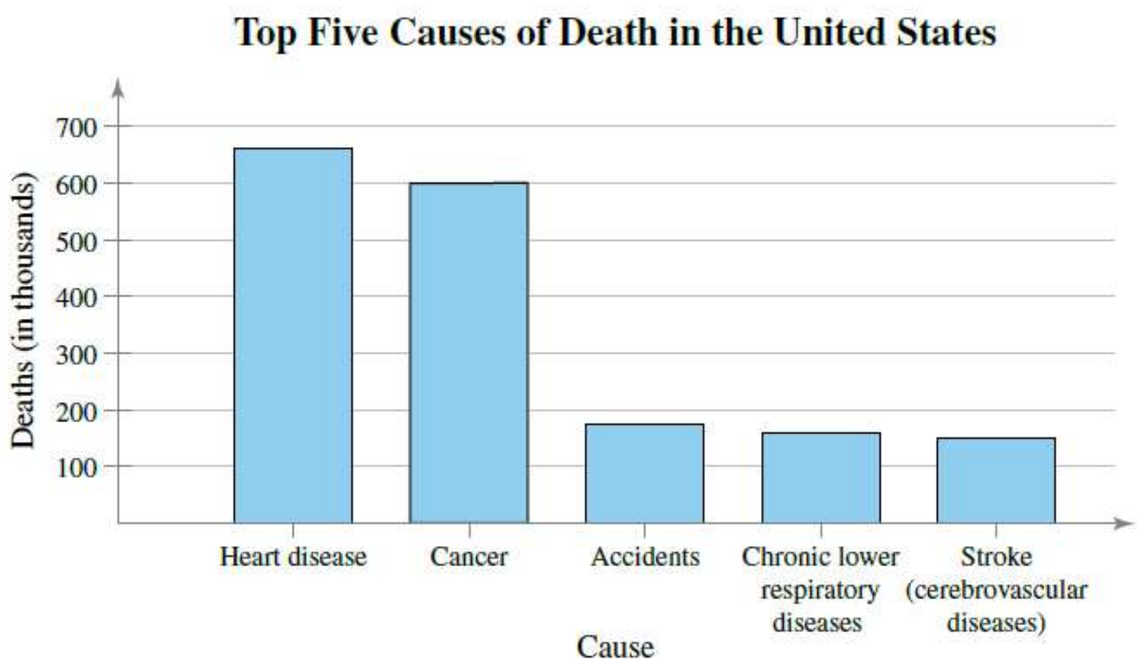
In 2019, these were the leading causes of death in the United States.

- Accidents: 173,040
- Cancer: 599,601
- Chronic lower respiratory disease: 156,979
- Heart disease: 659,041
- Stroke (cerebrovascular diseases): 150,005

Use a Pareto chart to organize the data. What was the leading cause of death in the United States in 2019?

(Source: National Center for Health Statistics)

Solution: Constructing a Pareto Chart

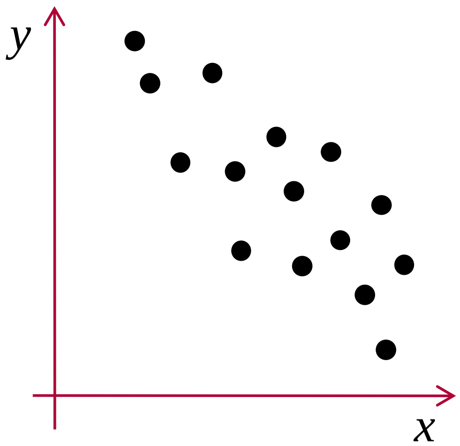


From the Pareto chart, you can see that the leading cause of death in the United States in 2019 was from heart disease. Also, heart disease and cancer caused more deaths than the other three causes combined.

Graphing Paired Data Sets

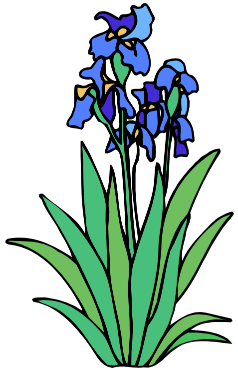
Paired Data Sets

- Each entry in one data set corresponds to one entry in a second data set.
- Graph using a **scatter plot.**
  - The ordered pairs are graphed as points in a coordinate plane.
  - Used to show the relationship between two quantitative variables.



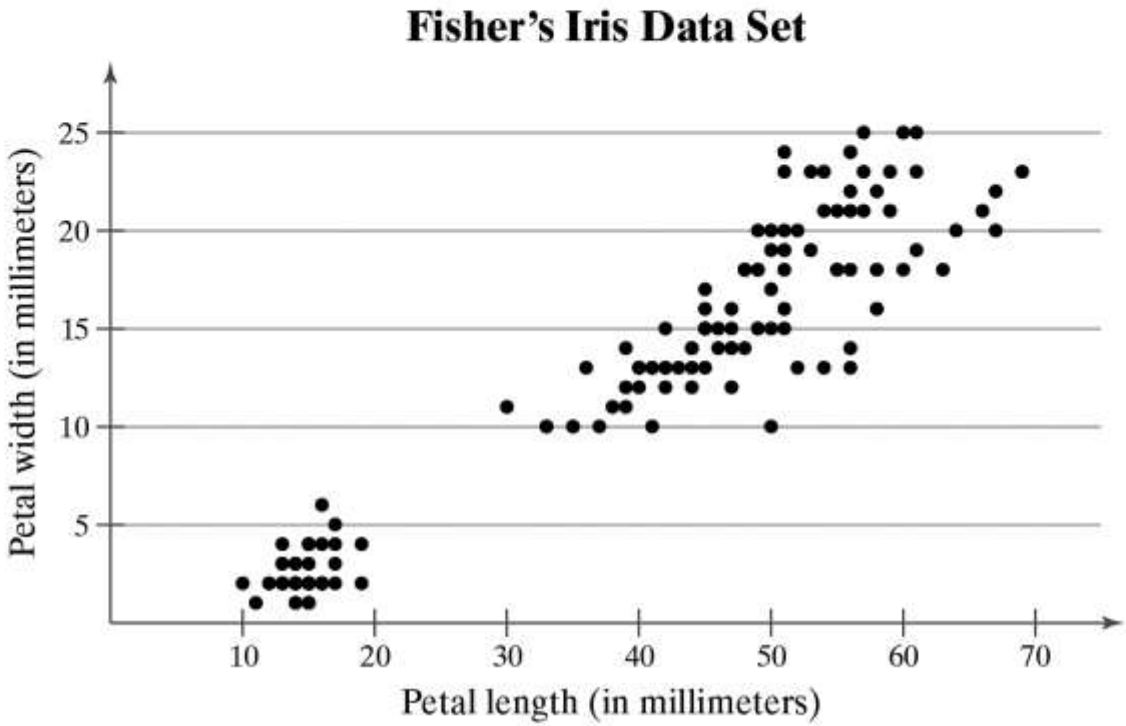
Example: Interpreting a Scatter Plot

The British statistician Ronald Fisher introduced a famous data set called Fisher's Iris data set. This data set describes various physical characteristics, such as petal length and petal width (in millimeters), for three species of iris. The petal lengths form the first data set and the petal widths form the second data set. *(Source: Fisher, R. A., 1936)*

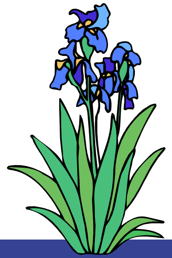


Example: Interpreting a Scatter Plot

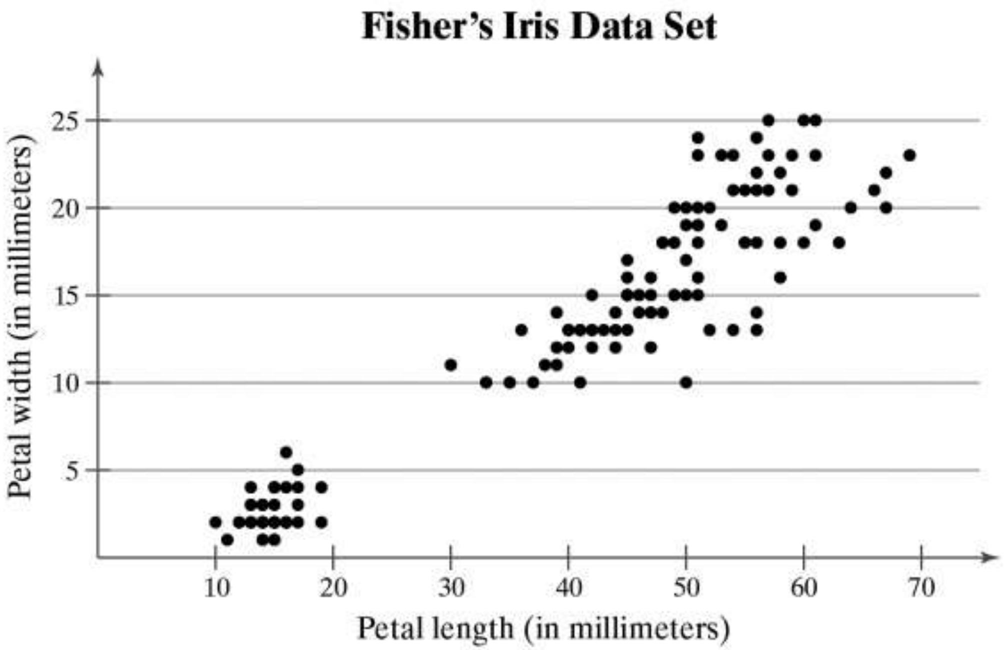
As the petal length increases, what tends to happen to the petal width?



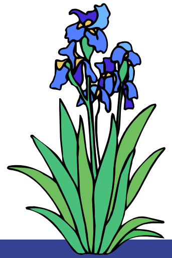
Each point in the scatter plot represents the petal length and petal width of one flower.



Solution: Interpreting a Scatter Plot



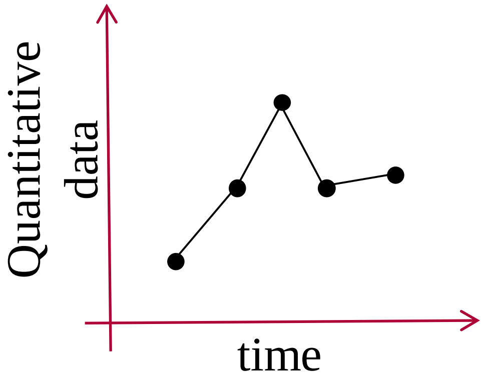
From the scatter plot, you can see that as the petal length increases, the petal width also tends to increase.



Graphing Paired Data Sets

Time Series

- Data set is composed of quantitative entries taken at regular intervals over a period of time.
  - e.g., The amount of precipitation measured each day for one month.
- Use a **time series chart** to graph.



Example: Constructing a Time Series Chart

The table lists the number of motor vehicle thefts (in millions) and burglaries (in millions) in the United States for the years 2009 through 2019. Construct a time series chart for the number of motor vehicle thefts. Describe any trends. *(Source: Federal Bureau of Investigation, Crime in the United States)*

Year	Burglaries (in millions)	Robberies (in thousands)
2009	2.20	409
2010	2.17	369
2011	2.19	355
2012	2.11	355
2013	1.93	345
2014	1.71	323
2015	1.59	328
2016	1.52	333
2017	1.40	321
2018	1.24	281
2019	1.12	268

Solution: Constructing a Time Series Chart

- Let the horizontal axis represent the years and let the vertical axis represent the number of motor vehicle thefts (in millions).
- Then plot the paired data and connect them with line segments.

Year	Burglaries (in millions)	Robberies (in thousands)
2009	2.20	409
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2013	1.93	345
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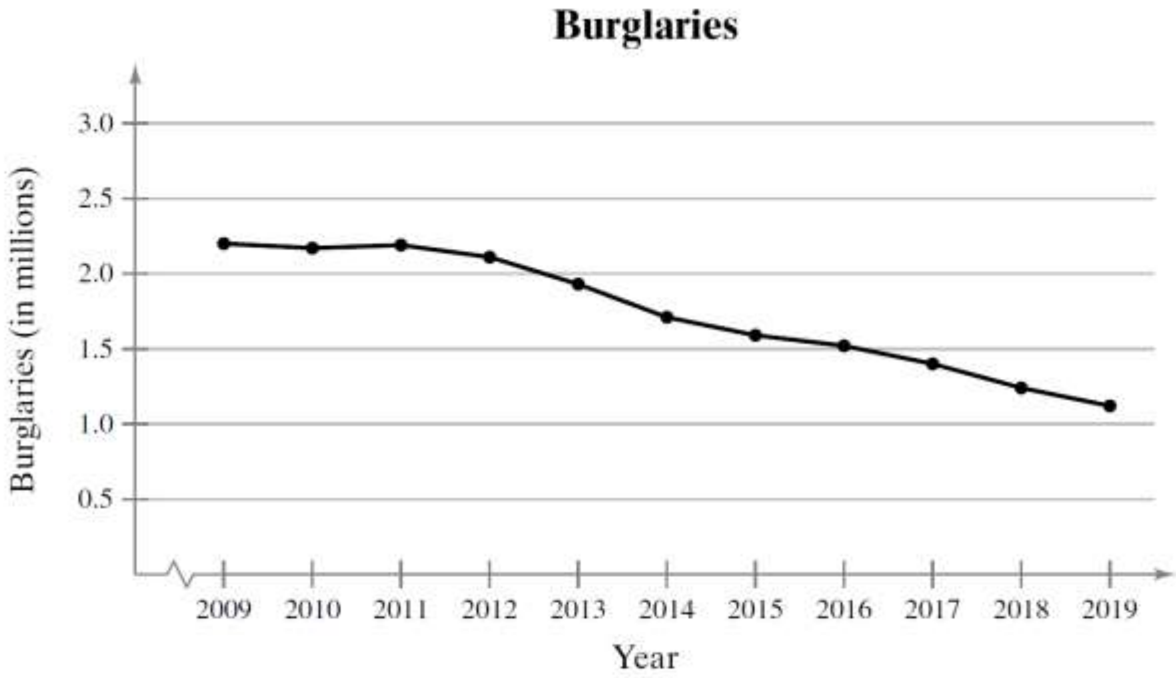
Solution: Constructing a Time Series Chart



The time series chart shows that the number of burglaries remained the same until 2011 and then decreased through 2019.

Exercise 1

Organize the data using the indicated type of graph. What can you conclude about the data?

Exam Scores Use a stem-and-leaf plot to display the data. The data represent the scores of a biology class on a midterm exam.

75	85	90	80	87	67	82	88	95	91	73	80
83	92	94	68	75	91	79	95	87	76	91	85

Exercise 2

Organize the data using the indicated type of graph. What can you conclude about the data?

United Nations Use a pie chart to display the data. The data represents the number of countries in the United Nations by continent.

North America	23	Europe	43	Africa	53
South America	12	Oceania	14	Asia	47

Exercise 3

Organize the data using the indicated type of graph. What can you conclude about the data?

Airline Baggage Use a Pareto chart to display the data. The data represents the results of 2005 worldwide study of all airlines on causes for baggage delay.

Transfer baggage mishandling	61%
Loading / offloading error	4%
Failure to load at originating airport	15%
Space-weight restriction	5%
Arrival station mishandling	3%
Other	12%

Exercise 4

Organize the data using the indicated type of graph. What can you conclude about the data?

UV Index Use a time series chart to display the data. The data represent the ultraviolet index for Memphis, TN, on June 14 – 23 during a recent year.

Date in June	14	15	16	17	18	19	20	21	22	23
UV index	9	4	10	10	10	10	10	10	9	9

Exercise 5

Customer Service Representatives

The back-to-back stem-and-leaf plot shows the number of hours worked per week (in hours) of all customer service representatives in two call centers.

Call Center A		Call Center B
9 8 7 6	2	3 4 5 7 9
9 8 7 6 5 3	3	0 1 2 3 5 6 8
9 8 7 6 5 4 2 1	4	1 2 3 4 5 6 7 9
8 7 5 4 2	5	0 1 2 3 4 6
9 7	6	2 4

Key: 3 | 3 | 0 = 33 hours for Call Center A and 30 for Call Center B

Exercise 5

1. What are the lowest and highest hours worked at Call Center A? at Call Center B?
2. How many representatives are in each call center?
3. Compare the distribution of hours worked at each call center. What do you notice?